

ASSESSMENT / 10 SEPTEMBER 2026

ZenPlus vs SolarWinds NPM

Network device monitoring: feature assessment and delivery plan

ZenPlus v1.23.10 | Development appliance | Revision 2: environment corrected

Revised assessment: development environment

Assess implemented features and confirmed gaps. Empty or simulated telemetry is expected here and is not a product defect. Real-device readiness was not tested.

Existing foundations include ICMP/SNMP collection, vendor templates, interface and hardware panels, NetPath, manual maps, alert configuration and report templates. Simulator records demonstrate UI and implementation surfaces, not real-device certification.

Development conditions are not product defects: no real SNMP devices, zero traffic, short history, offline test sensors and unconfigured channels are excluded from the feature-gap judgment. The previous revision gave these conditions excessive operational weight.

Remaining product gaps include topology lifecycle and routing depth, Nexus/ACI and cloud integrations, wireless client/RF visibility, F5 relationships, forecasting, dynamic baselines, a shared investigation timeline, compiled MIB tooling and syslog. Alert-evaluator source findings require reproduction; synthetic health discrepancies require a design/fixture review.

Plan: first confirm implementation gaps with deterministic fixtures and code tests, then build prioritized features. Real-device certification is a separate stage when equipment is available. The previous 8-12 / 20-28 week windows remain provisional scenarios pending this triage and test access, not revised commitments.

Capability status	Rows
Available / unproven	17
Observed	8
Unverified	4
Partial	25
Not found	17

71 assessed capabilities. Counts include explicitly marked platform, add-on and extension rows. They are not a parity percentage. Revision 2 supersedes the original operational-risk framing following the owner's clarification. See scope, findings, feature matrix, implementation plan, validation gates and evidence registers below.

1. Scope, evidence and interpretation

Revision 2 / user clarification: this is a development appliance without live SNMP or real network devices. The original review recognized simulator records but did not consistently apply that limitation. This revision supersedes its operational-risk framing; it does not add new runtime tests.

Live scope: existing administrator session on the requested appliance, observed on 10 September 2026 at approximately 15:27-15:41 Asia/Riyadh. Read dashboards, tables, detail views and unsaved editors. No device settings, rules, credentials, deployments, discovery scans, packet captures or outgoing test messages were changed or triggered.

Source corroboration: local C:/Users/user/Documents/ZenPlus checkout, .version 1.23.10 and HEAD b4256ff. The working tree contained unrelated modifications. Same version labels do not prove deployed binary/source identity; source findings are explicitly provisional for the appliance until deployment hash and runtime behavior are verified.

Comparator: full NPM functional scope and its shared SolarWinds Platform features, checked against current official documentation and NPM 2026.2 release notes. The directly opened documentation index advertised 2026.2.2; a search cache advertised 2026.2.1. This report uses feature documentation, not a claim that all patch-specific changes were validated. [S03 S04]

License boundaries: NTA, NCM, UDT, IPAM, VNQM and EOC are not silently counted as NPM. HA and additional polling engines have deployment/licensing dependencies. Newer fleet Routing Insights explicitly requires Observability Self-Hosted; it is labeled Extension. Validate current entitlements before procurement or a contractual parity claim. [S17 S26 S28]

Coverage: the matrix is an operational capability checklist across the documented NPM feature families, not a certification of every OID, supported device SKU, firmware release, integration action or deployment size. SolarWinds itself was assessed from official documentation, not a parallel live installation.

Evidence limits: synthetic or empty data cannot establish collector failure, missing functionality, production outages or poor availability. No fault injection, real-device polling, delivery or failover test was performed. Source-backed omissions and explicit deferred features are distinguished from implementation that simply awaits testing.

Status counts measure evidence categories, not completion or parity percentages. Available / unproven and Unverified are validation items, not missing-feature counts. Real-device accuracy, sustained reliability and scale are reserved for later certification.

Status definitions

Observed: The narrow feature was visible with current records or concrete controls. This is not an end-to-end parity certification.

Partial: Implemented with a specific feature-depth gap or source-backed semantic concern. Empty development data alone is not grounds for this status.

Available / unproven: An implementation or workflow is visible, but behavior has not been exercised. Unconfigured or empty development state is not a defect.

Not found: No implementation was located in the inspected UI and scoped source review, or the UI explicitly labels it deferred/planned. This is an evidence-bounded finding.

Unverified: The inspection cannot determine working support or its limits. A dedicated acceptance test is required.

Priority: P0 denotes a candidate defect requiring reproduction before acceptance; P1/P2 rank implementation depth. Validation means a test task, not a missing feature. A-H are workstreams; E is appliance UI evidence, C source evidence, S official comparator documentation.

2. Priority findings

Environment | Development conditions excluded from gaps

The user confirmed there are no live SNMP or real network devices on this appliance. Empty channels, short/synthetic history, zero throughput, no discovery runs and an offline test sensor are configuration/fixture conditions. They do not justify P0 product findings, a production outage claim or a restore-service workstream. Evidence E01-E04/E10/E13/E15.

P0 candidate | Compound alert evaluation

Local source C04 mirrors conditions[0] into legacy fields; C03 reads compound fields but the reviewed periodic branches evaluate the legacy metric. CPU >90 AND memory >90 could therefore use only the first condition. This concern is independent of real SNMP availability. Verify deployed identity and reproduce with injected metric truth tables before confirming a defect.

P0 candidate | Canonical state metric mapping

fan_state, psu_state, vpn_tunnel_state, ha_state and bgp_neighbor_down are accepted by the rule API but absent from the periodic canonical supported set. tpl_* is an alternative, so this is not absence of all component alerts. Trace accepted definitions through evaluation using fixtures; confirm or dismiss each mismatch. Evidence C03/C04.

P0 candidate | Hold and dependency semantics

Reviewed template evaluation uses latest values in a lookback, which does not establish a continuous breach. Periodic SNMP paths do not show the dependency checks used by event-driven alerts. Test hold/reset/no-data and parent-child cases with a controlled clock and injected state. Confirm deployed behavior before labeling appliance defects. Evidence C03/C05.

P1 review | Health score and stale demo semantics

Synthetic component faults alongside 100/100 and an old demo NetPath OK label are design/fixture questions. The score is described as CPU/memory/loss, so component faults need not imply a calculation bug. Agree labels, roll-up policy and demo behavior; reproduce using coherent enabled fixtures before requesting a fix. Evidence E02/E05/E08.

Priority findings / continued

Feature gaps | Explicit missing implementation remains relevant

Storage-only MIB uploads, absent exhaustion forecasts and the missing wireless-client/RF, cloud-controller, syslog and shared-timeline workflows do not depend on populated telemetry. Prioritize these based on source/UI evidence and target vendor scope. Automatic topology and vendor relationships have partial foundations rather than confirmed full coverage.

Validation | Real-device interoperability remains unknown

Existing collectors and vendor packs cannot be certified from UI examples. Later compare SNMPv2c/v3, counters, traps, interface identity, hardware and protocol states against representative hardware or vendor virtual appliances. Lack of such access is a test prerequisite, not evidence that the features are missing.

Validation | Reliability and scale require a separate stage

Coverage, notification delivery, collector continuity, access isolation, export accuracy, HA and capacity need dedicated tests. First use deterministic simulators and local test sinks; schedule real-device/scale qualification when available. These are future release criteria, not current development-appliance deficiencies.

3. Feature-by-feature assessment

Each row states the comparator capability, observed ZenPlus status, specific gap and recommended priority. "Not found" is limited to the inspected surfaces and scoped code review. A working control or lab reading does not certify production behavior.

Discovery and core polling		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F01 Network discovery and classification Network discovery NPM S02	Available / unproven Validation Workstream A	Discovery wizard and backend exist. Empty development history is expected; validate discovery lifecycle with fixtures, then a controlled subnet when available. Evidence: E04 C07
F02 Recurring discovery and import policy Scheduled discovery NPM S02	Available / unproven P1 Workstream B	Recurring/cron and review/auto-match implementation exists. No configured development profiles is not a feature gap; test scheduler and import policy with controlled fixtures. Evidence: E04 C07
F03 Resource selection and bulk management Manage nodes and resources NPM S02	Available / unproven P1 Workstream B	Device/interface selection, filtering and exports present; verify rediscovery preserves interface identity and monitoring choices. Evidence: E01 E03 E04
F04 ICMP status, RTT and packet loss Availability and response NPM S05	Observed Validation Workstream A	Status/history surfaces are present. Development availability values are not a production SLA result; later test missing-check and denominator policy. Evidence: E01 E02 E15
F05 SNMP v1/v2c/v3 polling SNMP polling NPM S05	Available / unproven P1 Workstream B	v2c simulator records and v3 model/session support exist. AuthPriv/context interoperability needs real-device or vendor-virtual-device certification; no failure inferred. Evidence: E02 E07 C01
F06 IPv6 network monitoring IPv6-capable monitoring NPM S02	Unverified P1 Workstream B	IPv6 discovery, ICMP, SNMP and tables require certification. Local NetPath is IPv4-only; this report does not assume SolarWinds NetPath supports IPv6. Evidence: C02 C06
F07 CPU, memory, uptime and identity Device performance NPM S01	Observed Validation Workstream A	Vendor and standard metrics appear on simulator records. Real-device numerical accuracy remains a separate certification task. Evidence: E02 E07 E08
F08 Polling intervals and diagnostics Configurable polling NPM S05	Available / unproven P1 Workstream B	Ping interval, SNMP diagnostics and bounded table cadence exist. Fleet poll completion, timeout causes and stale-resource health need validation. Evidence: E02 C01 C07

Discovery and core polling		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F09 Historical data and coverage Real-time and historical views NPM S01	Available / unproven Validation Workstream A	Historical charts and coverage disclosure exist. Short development history does not establish a retention or collection defect; test missing-data semantics and later sustained continuity. Evidence: E02 E15
F10 Tags, groups and custom properties Group and property scoping NPM S02	Partial P2 Workstream B	Tags/location/groups and tag-limited visibility exist. Arbitrary typed properties and dynamic group rules were not demonstrated. Evidence: E01 E16
Interfaces and device hardware		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F11 Interface inventory and status Interface monitoring NPM S02	Observed P1 Workstream B	47 links and per-device tables visible, including speed and status. Retain this foundation; validate reindex and administrative-down semantics. Evidence: E03
F12 Bandwidth and utilization Receive/transmit utilization NPM S01	Available / unproven Validation Workstream A	64-bit counters and rate handling exist. Zero/dashed development throughput is expected without traffic; verify calculations with synthetic counter sequences, then known device traffic. Evidence: E03 C01
F13 Errors and discards Interface health statistics NPM S02	Available / unproven P1 Workstream B	Fleet filters and reset-safe counter views exist; nonzero fault examples and rate/interval accuracy were not demonstrated. Evidence: E03 C01
F14 Link flaps and downtime Interface outage diagnostics NPM S02	Available / unproven P1 Workstream B	Flapping-link counters and detail views present. Test up/down sequences, missing samples and per-link timelines. Evidence: E03
F15 Custom interface capacity Per-interface configuration NPM S02	Observed P1 Workstream B	Manual speed/bandwidth control visible. Prove asymmetric circuit calculations and direction labels. Evidence: E03
F16 Duplex mismatch diagnostics Detect/predict mismatch NPM S02	Not found P2 Workstream D	No dedicated duplex state or mismatch diagnosis located. Add polling, peer correlation and actionable alerts for supported media. Evidence: E03 C01
F17 Physical device / port layout Device View stencils NPM S02	Not found P2 Workstream D	Interface tables are present; chassis/slot/port stencils with live status were not found. Evidence: E02 E03

Interfaces and device hardware		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F18 Temperature, fans and PSU Hardware health NPM S27	Partial P1 review Workstream A	Hardware panels exist. Synthetic faults alongside a 100/100 performance score raise a roll-up/labeling design question, not proof of failed physical sensors. Reproduce with coherent fixtures and agree score scope. Evidence: E02 E07 E08
F19 Switch stacks and redundancy Stack monitoring NPM S02	Partial P1 Workstream D	Cisco members/ring observed; CX VSF pack exists. Juniper Virtual Chassis and complete cross-vendor stack health not demonstrated. Evidence: E02 E07 C11
F20 PoE, optics and modular hardware Device-specific hardware NPM S04	Partial P2 Workstream D	PoE visible and MikroTik optics defined. Broader optics, line-card state, reset reasons and inventory relationships need coverage. Evidence: E02 E07 C11
Topology and network protocols		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F21 Manual maps and NOC display Maps and NOC views NPM S01	Observed P1 Workstream B	Rich editable map with live mode and interface labels exists. The inspected map was QA/simulator data. Evidence: E06
F22 LLDP/CDP topology collection Topology relationships NPM S01	Available / unproven P1 Workstream D	Quick-connect consumes discovered adjacency. Validate refresh, stale links, port resolution and multi-vendor discovery. Evidence: E06 C09
F23 Automatic topology lifecycle Dynamic topology mapping NPM S01	Partial P1 Workstream D	Manual canvas assistance is not proof of automatically maintained topology. Add change history, aging and confidence for discovered links. Evidence: E06 C09
F24 Dependencies and impact isolation Dependency-aware alerting NPM S01	Partial P0 Workstream C	Event-driven suppression exists; periodic SNMP parity and an operator-facing dependency workflow were not established. Evidence: E09 C05 C03
F25 BGP and OSPF neighbor health Routing neighbor visibility NPM S02	Partial P1 Workstream D	BGP/OSPF tables exist in collector; no complete routing workflow demonstrated. Canonical BGP alert key needs evaluator mapping. Evidence: E07 C02

Topology and network protocols		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F26 Routing tables and route changes Route visibility NPM S02 S17	Not found P1 Workstream D	Dedicated route inventory, next-hop correlation and route/default-route change views not located. Scope original NPM versus newer Insights separately. Evidence: C02 E06
F27 VRRP, STP and LACP health Network protocol context NPM S02	Partial P1 Workstream D	Standard tables implemented. Dedicated protocol relationships, transitions and object-level alert lifecycle still need verification. Evidence: C02
F28 Multicast groups and routing L3 multicast monitoring NPM S16	Not found P2 Workstream D	No multicast routing/group collector or tree workflow found. Flow protocol classification is not multicast routing monitoring. Evidence: E06 C02
F29 Fleet routing insights, VRFs and peer correlation Unified Routing Insights Extension S17 S18	Not found P2 Workstream D	Candidate extension: no unified VRF/peer/route interface was found. Official fleet Routing Insights requires Observability Self-Hosted licensing. Evidence: C02 E06
Vendor, wireless and hybrid coverage		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F30 Cisco ASA insight ASA Network Insight NPM S27	Partial P1 Workstream E	ASA pack includes 19 metrics for connections, failover and remote-access counts. Zero attached development devices is not a gap; required per-tunnel/user drill-down depth remains to be established. Evidence: E07
F31 Cisco Nexus vPC insight Nexus Network Insight NPM S11	Not found P1 Workstream E	No dedicated NX-OS/vPC collector or workflow located. IOS/XE metrics do not establish Nexus peer/keepalive/member-port parity. Evidence: E07 C11
F32 Cisco ACI logical / physical context ACI monitoring NPM S27	Not found P2 Workstream F	No APIC integration or tenant/application-profile/EPG/fabric workflow found. Build only against a representative fabric and fixed scope. Evidence: E07
F33 F5 LTM component monitoring F5 virtual servers and pools NPM S12 S13	Partial P1 Workstream E	39-metric SNMP pack provides virtual-server/pool health, traffic, HA and hardware. Relationship graph and pool-member-level parity not demonstrated. Evidence: E07

Vendor, wireless and hybrid coverage		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F34 F5 DNS/GTM and health monitors GTM and iControl NPM S12	Not found P2 Workstream E	No GTM/Wide-IP hierarchy or iControl health-monitor workflow located; optional pool-member rotation controls also absent. Evidence: E07
F35 Palo Alto firewall insight Palo Alto Network Insight NPM S04 S02	Partial P1 Workstream E	PAN-OS pack defines sessions, CPU, HA and GlobalProtect. Complete site-to-site tunnel state/traffic/encryption detail is not established. Evidence: E07
F36 FortiGate monitoring Fortinet network monitoring NPM S04 S08	Available / unproven P1 Workstream E	Rich FortiGate SNMP panels and template implementation exist. Physical or virtual FortiOS validation is pending; simulator-only data does not reduce implemented coverage. Evidence: E08
F37 Juniper, Aruba, Dell, MikroTik and other vendors Multi-vendor device coverage NPM S02 S04	Partial P2 Workstream E	Several vendor packs exist. Unassigned packs are not missing. Broader Arista/Extreme support was not established; assess family depth separately from model/firmware certification. Evidence: E07 C11
F38 Wireless controller and AP inventory Wireless monitoring NPM S08	Partial P1 Workstream E	Aruba/FortiGate AP and client totals exist. No unified wireless inventory or Cisco WLC monitoring pack demonstrated. Evidence: E07 E08
F39 Per-client wireless performance Client, SSID and radio detail NPM S10	Not found P1 Workstream E	No client-level RSSI/SSID/radio/channel history located. Aggregate client counts and UDT endpoint records do not meet this capability. Evidence: E07 E08
F40 Rogue APs and RF coverage maps Wireless diagnostics NPM S08 S09	Not found P2 Workstream E	No rogue-AP workflow or RF heat map found. SolarWinds heat maps have Cisco-controller restrictions and legacy Network Atlas dependencies. Evidence: E06 E07
F41 Cloud-managed wireless APIs Cloud controller monitoring NPM S08 S10	Not found P1 Workstream F	Meraki, Aruba Central, Mist, Ruckus and similar API connectors not found. Add vendor API pagination, rate limits and health diagnosis. Evidence: E07
F42 SD-WAN orchestrators and overlay Orchestrator/edge/uplink monitoring NPM S14 S04	Partial P1 Workstream F	FortiGate SNMP WAN health exists. No VeloCloud/Catalyst/Prisma/FortiManager orchestration integration or overlay inventory found. Evidence: E08

Vendor, wireless and hybrid coverage		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F43 Azure gateways and VPN connections Azure network monitoring NPM S15	Not found P2 Workstream F	No Azure gateway/site-to-site account integration found. NPM scope includes gateways/connections, not all cloud observability. Evidence: E07
Troubleshooting and customization		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F44 Continuous hop-by-hop path monitoring NetPath NPM S19	Observed P1 Workstream B	Live TCP SYN probe, multiple flows, latency/loss and branching path present. Retain and validate path semantics and target reachability. Evidence: E05
F45 Path history and route comparison Historical NetPath NPM S19	Observed P0 Workstream A	History, routes, events and compare controls plus endpoints exist. A stale demo OK label is a fixture/freshness-policy question; reproduce on an enabled test probe before calling it a defect. Evidence: E05 C06
F46 Multiple network-path vantage points Distributed path probes NPM S19 S26	Unverified P1 Workstream F	Multiple-vantage NetPath execution was not verified. Offline development sensors do not establish lack of support; inspect capability wiring and test an isolated remote probe. Evidence: E05 E13 C06
F47 Shared cross-metric timeline PerfStack NPM S19	Not found P1 Workstream G	Individual charts exist; no saved shared-timeline investigation across devices, interfaces, paths, alerts and applications located. Evidence: E02 E05 E14
F48 Custom OID scalar/table polling Universal Device Pollers NPM S06	Available / unproven P1 Workstream B	Template groups, counters, enums and scale are implemented. Validate arbitrary custom OIDs, row identity and reusable import/export workflows. Evidence: E07 C01
F49 MIB resolution and derived formulas MIB and poller customization NPM S06 S07	Partial P1 Workstream B	MIB upload is storage-only. No complete symbolic MIB browser/compile path or arbitrary multi-poller formula editor demonstrated. Evidence: E12 C01
F50 Network capacity forecasts Exhaustion-date prediction NPM S20	Not found P1 Workstream G	Headroom and 95th percentile exist, but no CPU/memory/interface exhaustion forecast found. Add trend models and data-quality gates. Evidence: E17 C10

Troubleshooting and customization		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F51 Dynamic statistical thresholds Baseline thresholds NPM S21	Not found P1 Workstream G	Static thresholds are present. Server software baselines are a different feature; network baseline learning was not found. Evidence: E09 C03
F52 Passive application/network timing Quality of Experience NPM S24 S25	Partial P2 Workstream F	On-demand connection capture exists in source. No continuous SPAN/server sensor pipeline showing network versus application response time established. Evidence: C13
Alerting, logs and reporting		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F53 Threshold rules and recovery Advanced alerting NPM S22	Partial P0 Workstream C	Core scalar/interface rules exist; verify all supported metrics, clear conditions and mixed-engine behavior before operational sign-off. Evidence: E09 C03
F54 AND/OR compound conditions Multi-condition alerts NPM S01	Partial P0 Workstream C	Wizard exposes AND/OR. Reviewed periodic network evaluator selects legacy metric fields and does not apply the conditions array. Deployment verification required. Evidence: E09 C03 C04
F55 Hold time and reset hysteresis Sustained thresholds NPM S22	Partial P0 Workstream C	Scalar sustained logic exists; template rules evaluate latest values from a lookback. Complete reset/hold semantics across every evaluator. Evidence: E09 C03
F56 Component-specific alert lifecycle Hardware/protocol alerts NPM S13 S27	Partial P0 Workstream C	Canonical fan/PSU/VPN/HA/BGP keys are accepted but absent from periodic supported set. tpl_* provides an alternative, with device-level aggregation. Evidence: E07 E09 C03 C04
F57 Notifications and escalation Alert actions and escalation NPM S27	Available / unproven Validation Workstream A	Channel configuration, cooldowns and escalation controls exist. Zero destinations and disabled development gateways are expected configuration state. Test transport with local sinks before optional real delivery. Evidence: E09 E10
F58 Maintenance, snooze and dependencies Alert suppression NPM S04	Partial P0 Workstream C	Maintenance/snooze controls exist. Prove node-down, dependency, schedule and maintenance suppression consistently across metric and event engines. Evidence: E02 E09 C03 C05

Alerting, logs and reporting		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F59 ITSM and collaboration workflows Platform integrations Platform S27	Partial P2 Workstream G	Webhook/Slack/Telegram cataloged ready; Teams and Jira/ServiceNow explicitly planned. Generic webhook does not establish bidirectional incident sync. Evidence: E10
F60 SNMP trap reception and history Trap ingestion NPM S23	Available / unproven P1 Workstream C	v1/v2c receiver and feed exist; an empty development feed is not failure. v3 traps are explicitly deferred in source, independently of test traffic; verify required comparator scope before prioritizing. Evidence: E11 C08
F61 Network-device syslog workflow Log Viewer NPM S23	Not found P1 Workstream C	No network syslog receiver/search/rule workflow found. Server logs and a Syslog port label do not establish ingestion. Evidence: E11 C08
F62 Alert/event history and acknowledgement Operational alert console NPM S27	Observed P1 Workstream C	Live events, active alerts and acknowledge controls visible. Alert storm, grouping, stale-state and operator workflow tests remain. Evidence: E01 E02 E09
F63 Availability/SLA and outage reports Availability reporting NPM S02	Available / unproven Validation Workstream A	Availability views, report templates and coverage disclosure exist. Short synthetic history is not a product gap; validate denominator, maintenance and unknown-data policy with fixtures, then a real-device pilot. Evidence: E14 E15
F64 Custom reports and scheduled export Report builder and schedules NPM S02	Partial P1 Workstream G	Fixed-section reporting and export/schedule controls exist. Arbitrary object/field/query authoring was not found in the inspected builder. Empty schedules/history are development configuration, not missing capability. Evidence: E14
Platform and enterprise readiness		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F65 Role and object access controls Roles and scoped access Platform S27	Available / unproven P1 Workstream H	Viewer tag scope and permission model visible. Read-only inspection cannot prove enforcement on API, reports, exports and nested objects. Evidence: E16 C12
F66 Directory authentication and SSO Enterprise authentication Platform S27	Partial P2 Workstream H	LDAP/RADIUS implementation exists; lack of configured development authentication is not a defect. SAML/OIDC support was not established; track this scope separately from integration testing. Evidence: E16 C12

Platform and enterprise readiness		
Feature / SolarWinds baseline	Status / priority	ZenPlus evidence and required work
F67 API and auditability Platform extensibility Platform S27	Partial P2 Workstream H	FastAPI and audit logging exist in source. Public integration contract, scoped API lifecycle and complete audit coverage need certification. Evidence: C04 C06 C12
F68 Distributed polling and buffering Remote collectors Platform S26	Available / unproven Validation Workstream A	Remote sensor deployment and assignments exist. An offline development sensor is not a distributed-polling defect; verify buffering, replay, timestamps and version compatibility in an isolated test. Evidence: E13
F69 Polling capacity and load distribution Additional polling engines Add-on S26	Unverified P1 Workstream H	40-device mixed estate is not a scale benchmark. No measured sustainable element capacity, engine balancing or saturation behavior established. Evidence: E01 E13
F70 Monitoring-system high availability HA deployment Add-on S04 S27	Not found P1 Workstream H	No controller/collector failover pools or proven database failover found. Device HA panels monitor equipment; they do not protect ZenPlus itself. Evidence: E13
F71 Retention, backup and restore Operational continuity Platform S26	Unverified P1 Workstream H	Settings exist in navigation; historical continuity, restore integrity and disaster recovery were not exercised. Define and prove RPO/RTO. Evidence: E14 E15

4. Keep the module boundaries clear

A full NPM review must include its shared platform functions, while separating other products. SolarWinds documents NPM, NTA, UDT, VNQM, NCM and IPAM as distinct NAM components. Configuration compliance, firmware and approval workflows belong to NCM. [S28 S29]

Adjacent capability	SolarWinds scope	Treatment in this assessment
NetFlow traffic analytics	NTA	ZenPlus has NetFlow overview, forensics, anomalies and capacity screens; the inspected windows contained no flows. Treat as an adjacent capability, not proof of NPM depth or NTA parity.
Configuration backup/compliance/firmware	NCM	Config Backup is present in ZenPlus navigation. A backup menu does not establish configuration compliance, approval, vulnerability, rollback or firmware-upgrade parity.
Endpoint/MAC/user/switch-port tracking	UDT	ZenPlus has an extensive UDT navigation surface and related collector code. A full UDT assessment was outside this NPM review.
IP address and DHCP/DNS management	IPAM	Do not mark IPAM absence as an NPM deficiency. Assess separately if network-management-suite replacement is intended.
VoIP quality and IP SLA operations	VNQM	Do not count full voice/call-manager/IP-SLA coverage as an NPM requirement without expanding scope.
Deep server/application/virtualization observability	SAM / VMAN / other	ZenPlus APM and server modules extend beyond this review. They do not compensate for missing network-device functions.
Multiple-instance enterprise console	EOC	Separate federation requirement; a single controller plus remote sensors is not evidence of a multi-instance enterprise console.

Licensing and legacy caveats

Fleet Routing Insights is explicitly an Observability Self-Hosted extension. HA and scalability engines require deployment/entitlement review. Cisco RF heat maps depend on supported Cisco controllers and legacy Network Atlas; current release notes deprecate Network Atlas. Do not prioritize copying a legacy UI without checking its successor and your RF requirement. [S17 S26 S09 S04]

5. Implementation plan

Recommended sequence

Start with development fixtures and reproduction of source candidates. Implement confirmed gaps in alerting, MIBs, topology and required device workflows, then analytics and integrations. Schedule real-device interoperability, multi-site continuity and scale certification separately when infrastructure is available. Workstream windows below are provisional sequencing scenarios to re-estimate after triage.

Revision 2 separates development work from real-device certification. Weeks 1-2 establish deterministic fixtures and reproduce source candidates; there is no instruction to repair an idle development sensor or configure production channels as a product fix.

Previous estimates of 80-129 person-weeks, 8-12 weeks for qualified core scope and 20-28 weeks for broad enterprise scope are provisional scenarios, not validated revised estimates. Workstream A has changed scope. Re-estimate after fixture-based triage; untested behavior must not automatically become implementation work.

The earlier calendar scenario assumed about five effective delivery engineers plus part-time product/network ownership and timely access to equipment/APIs. Missing access leaves real-device certification unscheduled; the development calendar must not imply that certification has already been possible.

Use two tracks: implement source/UI-backed missing capabilities and reproduce candidate defects now; run model/firmware, delivery, multi-site, soak and recovery qualification when appropriate hardware, vendor virtual appliances and external systems are available.

The later real-device pilot may target 20 representative nodes across two sites, 100 interfaces and 14 days of evidence, followed by scale/soak testing. These are proposed acceptance targets to agree, not prerequisites for this development review or observed current deficiencies.

Keep NPM requirements, shared platform functions and separately licensed modules distinct. Prioritize target device families, versioned schemas and reversible releases. Confirm numerical accuracy and vendor compatibility separately from simulator-based semantic correctness.

Week-1 decisions to finalize scope

Which vendors, models and firmware are launch requirements, and which can be tested now using hardware, vendor virtual appliances or simulators?

What are the target nodes, interfaces, wireless clients, traps/syslogs per second, polling intervals and retention? Device count alone is insufficient for sizing.

Is the commercial target standalone NPM replacement or the broader NAM/Observability suite? Keep adjacent modules and licensed extensions in separate acceptance scopes.

Which fixtures, mock notification sinks and test accounts can support development validation, and when can a separate real-device pilot begin?

Workstream summary

Workstream	Calendar window	Effort
A - Build development validation and confirm findings	Weeks 1-2	4-6 person-weeks
B - Harden discovery and polling contracts	Weeks 2-6	6-9 person-weeks
C - Unify alert semantics and network events	Weeks 1-6	8-12 person-weeks
D - Complete topology and protocol workflows	Weeks 5-10	8-12 person-weeks
E - Certify vendor insight and wireless depth	Weeks 7-16	14-22 person-weeks
F - Add hybrid and distributed diagnostics	Weeks 11-22	12-20 person-weeks
G - Forecasting, investigations and reporting	Weeks 7-14	8-12 person-weeks
H - Enterprise reliability and release gates	Weeks 9-24; certification buffer to week 28	10-16 person-weeks

A | Build development validation and confirm findings

Weeks 1-2 | 4-6 person-weeks | Owner: SRE + backend + QA

Dependencies: None

Deliver: Map deployed build to source; prepare deterministic SNMP/counter fixtures, a controlled test clock, local notification sinks and remote-probe fixtures. Reproduce alert candidates and agree health/freshness semantics. Catalogue implemented, missing and unverified behavior; size confirmed defects only.

Acceptance (A): Repeatable scenarios cover known counter rates/resets, missing data, compound alerts, hold/recovery and simulated component faults. Results distinguish fixture behavior from product defects. No physical network equipment or production destination is required for this stage.

Primary feature rows: F01, F04, F07, F09, F12, F18, F45, F57, F63, F68

B | Harden discovery and polling contracts

Weeks 2-6 | 6-9 person-weeks | Owner: Collector + backend + QA

Dependencies: A: fixtures and representative model/firmware test plan

Deliver: Verify discovery/polling contracts with synthetic records; schedule real-device SNMPv3 and IPv6 interoperability separately. Test selection/reindex, wrap/reboot/discontinuity and freshness behavior. Implement missing MIB resolve/compile and custom-poller reuse workflows after checking existing depth.

Acceptance (B): Discover a controlled subnet twice without duplicate identities. Preserve intended monitoring after ifIndex changes. Certify v2c/v3, 1/10/100-Gbps counters as applicable, resets and missing samples against device/traffic-generator reference. Custom scalar/table and transformed metric are usable in charts and alerts.

Primary feature rows: F02, F03, F05, F06, F08, F10, F11, F13, F14, F15, F21, F44, F48, F49

C | Unify alert semantics and network events

Weeks 1-6 | 8-12 person-weeks | Owner: Backend lead + collector + QA

Dependencies: A fixtures; code tracing and unit-level reproduction can start immediately

Deliver: Confirm the source candidates before changing code, then correct reproduced AND/OR, hold/reset, dependency and component-identity defects. Add capability validation, syslog workflow and required trap support. Exercise transports with local sinks during development.

Acceptance (C): Controlled tests cover supported metric families, parent-child suppression, independent tunnel state, hold/reset and replay idempotence. Synthetic traps/syslog reach local notification sinks. Real-device protocol and external-channel certification remains a later gate.

Primary feature rows: F24, F53, F54, F55, F56, F58, F60, F61, F62

D | Complete topology and protocol workflows

Weeks 5-10 | 8-12 person-weeks | Owner: Collector + UI + QA

Dependencies: B identity/counters; C object alerting

Deliver: Discovered topology with aging and change history; explicit dependencies; BGP/OSPF route/neighbor detail; STP/LACP/VRRP relationships; duplex diagnostics and stack depth. Stage multicast and physical device views by demand.

Acceptance (D): Compare a two-site reference topology with LLDP/CDP and device CLI. Link removals age correctly; routed next hops and neighbors map correctly. Simulated peer/stack/link failures show impact and suppress child noise. Multicast/VRF extensions have separate acceptance scope.

Primary feature rows: F16, F17, F19, F20, F22, F23, F25, F26, F27, F28, F29

E | Certify vendor insight and wireless depth

Weeks 7-16 | 14-22 person-weeks | Owner: Two collector/integration engineers + UI + QA

Dependencies: B custom metrics; C component alerts; D relationships

Deliver: Prioritize installed device families. Add Nexus vPC, complete ASA/PAN-OS tunnels, F5 pool members/relationships and optional GTM/iControl; implement Cisco WLC plus wireless clients/SSID/radio history. Expand hardware/stack coverage with a model/firmware matrix.

Acceptance (E): For each supported family use at least two representative firmware/model combinations where feasible, plus a documented negative/unsupported case. Compare counters and states with vendor output. Prove component failure/recovery and no stale health. Wireless client identity and RF metrics match controller data.

Primary feature rows: F30, F31, F33, F34, F35, F36, F37, F38, F39, F40

F | Add hybrid and distributed diagnostics

Weeks 11-22 | 12-20 person-weeks | Owner: Integration + collector + UI + QA

Dependencies: B/C/D; actual controller/API access

Deliver: Select one SD-WAN and one cloud-wireless provider first; then expand. Add Azure gateway integration and ACI if deployed. Certify remote NetPath. Design continuous packet-based QoE as a separately sized workstream.

Acceptance (F): Controller entities reconcile without duplicates under pagination, credential expiry and API throttling. Tunnel uplink metrics match vendor console. Same target measured from two sites retains correct source identity. QoE, if selected, separates network and application timing under controlled induced latency.

Primary feature rows: F32, F41, F42, F43, F46, F52

G | Forecasting, investigations and reporting

Weeks 7-14 | 8-12 person-weeks | Owner: Backend/data + UI + QA

Dependencies: A synthetic history; B metric metadata; C events; real history for later forecast qualification

Deliver: Shared time-aligned investigation workspace; saved/shareable read-only views; baseline thresholds; interface/CPU/memory forecasts; richer object-field reports; scheduled delivery. ITSM integration is optional scope.

Acceptance (G): Overlay interface, device, path and alert data at one timestamp. Forecasts use peak/average trends, require sufficient history and suppress misleading dates on flat/noisy/sparse data. Backtest on held-out periods. Report and exported data match source queries and preserve access limits.

Primary feature rows: F47, F50, F51, F59, F64

H | Enterprise reliability and release gates

Weeks 9-24; certification buffer to week 28 | 10-16 person-weeks | Owner: SRE + platform/backend + QA/security

Dependencies: A-C verified; agreed target scale and later test infrastructure

Deliver: Define supported element capacity; load and soak testing; HA/DR architecture and deployment automation; collector reassignment; backup/restore; RBAC/API/export isolation; authenticated integration documentation.

Acceptance (H): Run 72-hour load tests at agreed target scale and a 30-day pilot. Proposed starting target: 1,000 nodes / 10,000 interfaces, validated at chosen polling intervals rather than claimed in advance. Demonstrate controller failover with no duplicate actions, RTO ≤ 5 min and RPO ≤ 1 min if architecture supports them; otherwise publish measured limits.

Primary feature rows: F65, F66, F67, F69, F70, F71

6. Validation and release decisions

These are proposed tests, not tests completed during this assessment. Run fault injection in an isolated test network or approved maintenance environment. Replace every "unverified" claim with reproducible evidence before making a parity claim.

Test	Required proof	Owner lane
T01 Polling correctness	Compare expected polls with persisted timestamps; v2c/v3, timeouts and IPv6; prove no-data differs from zero or down.	A/B
T02 Traffic accuracy	Known traffic at several rates; counter wrap, restart, link-speed change and ifIndex change; direction-specific utilization.	B
T03 Discovery lifecycle	Repeated scan, new interface, removed device, changed IP/hostname, duplicates, exclusions and import audit.	B
T04 Alert semantic contract	AND/OR truth tables, hold, reset, recovery, maintenance, schedules and all canonical/template state metrics.	C
T05 Fault impact and notification	Parent failure suppresses children; independently failing tunnels; message retries and replay; acknowledgement stops escalation.	A/C/D
T06 Vendor correctness	Model/firmware matrix with real reference output; unsupported OIDs explicit; sensor alarms propagate to summary.	D/E
T07 Wireless and controller lifecycle	Roaming/disconnect, client/SSID/radio association, API pagination/rate limits/credential expiry, stale children.	E/F
T08 Path correctness	Multi-flow branch behavior, target timeout versus intermediate ICMP silence, history comparison, stale expiry and source identity.	B/F
T09 Analytics and reports	Coverage-aware SLA, forecast backtesting, synchronized timeline, exported values and scheduled-channel delivery.	G
T10 Enterprise recovery and access	72-hour scale test, 30-day soak, collector outage replay, controller/database failure, restore and cross-role API/export isolation.	H

Release gates

Gate 1 - Development validation: deterministic fixtures reproduce or dismiss source candidates; counters, alert semantics, missing-data policy and local notification sinks pass; health and demo labels match agreed behavior.

Gate 2 - Real-device qualification, when equipment is available: discovery/counters and required router/switch/firewall families are certified; known traffic, faults, delivery and topology impact match reference outputs; no confirmed critical defects remain.

Gate 3 - Enterprise release: required wireless/controller integrations, forecasting/reporting, scale, access isolation and recovery objectives pass a sustained pilot. Publish the supported-feature/device/firmware matrix and explicit limits.

Full parity claim: all required baseline rows are evidenced at the agreed deployment scale. Exclusions, optional licenses, legacy features and customer-specific integrations must be explicit in the acceptance agreement.

7. Appliance evidence register

Links reopen the live appliance and may show a different state later. This register records the assessment observations, not immutable screenshots. No secret values are included.

E01 | Dashboard and inventory

[Open appliance view](#)

40 monitored devices; the snapshot mixed lab, QA, private-network and Internet targets. Live alert totals changed during inspection. Do not interpret these totals as a production incident census. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E02 | Cisco device detail

[Open appliance view](#)

cisco-lab-01, 127.0.0.15, SNMP v2c on port 16161. CPU, memory, temperature, fans, PSU, stack and PoE visible. Fan warning coexisted with a 100/100 score. One-hour availability initially covered 15% of the range. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E03 | Interfaces and link utilization

[Open appliance view](#)

47 monitored links, zero with NetFlow; displayed throughput was zero/dashes and fleet utilization 0%. Filters, errors, discards, flaps, drill-down and manual interface capacity controls were visible. Meaningful loaded-link measurements were not demonstrated. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E04 | Discovery

[Open appliance view](#)

No discovery profiles; no import history in /discovery/imports. Wizard exposes scope, credentials/protocols, schedule, classification/rules and review. No scan, import or saved draft was started. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E05 | NetPath list and live detail

[Open appliance view](#)

Five probes; a live HTTPS path had TCP SYN, four flows, history, branching hops, ASN ownership, routes, events and comparison controls. A separately labeled demo probe retained an OK label with a last check 28 days ago. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E06 | Network Studio

[Open appliance view](#)

Selected Traffic lab (QA) map described scratch simulator devices: seven devices, eight links, three shapes. Manual design, live mode, NOC fullscreen, interface labels and LLDP/CDP quick-connect control were present; automatic lifecycle maintenance was not demonstrated.

E07 | Monitoring templates

[Open appliance view](#)

10 built-in packs, 222 metrics, eight attached devices. Packs: Aruba CX, Aruba controller, Cisco ASA, Cisco IOS/XE, Dell OS10, F5, FortiGate, JunOS, MikroTik and PAN-OS. ASA, CX, Dell and MikroTik had zero attached devices. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E08 | FortiGate detail

[Open appliance view](#)

fgt-lab-01, 127.0.0.14, SNMP v2c on port 16161. VDOM, HA, VPN, SD-WAN, FortiAP and FortiSwitch panels populated. A 100/100 score coexisted with an unsynchronized HA member, down tunnel, dead WAN check and PSU alarm. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E09 | Alert definitions and wizard

[Open appliance view](#)

Existing ICMP, SNMP interface/CPU/memory/reboot and other rules. Nine-step wizard exposes AND/OR conditions, hold duration, scope, reset, trigger actions, escalation and schedules. Inspected without saving or sending tests.

E10 | Notification channels

[Open appliance view](#)

Zero configured channels; two default SMTP/SMS gateways disabled; alert routing not set. Catalog labels Email, SMS, Webhook, Slack and Telegram ready; Teams and Jira/ServiceNow planned. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E11 | Trap receiver UI

[Open appliance view](#)

UDP/162 feed with time/severity filters; zero traps in selected one-hour window. This is an empty observation, not proof of receiver failure. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E12 | MIB library

[Open appliance view](#)

Zero installed MIBs. UI explicitly states uploaded files are stored on disk and runtime compilation will arrive in a later update.

E13 | Remote sensor fleet

[Open appliance view](#)

One VMware sensor, v1.23.5, seven assignments, heartbeat approximately two days old, offline. OVA/OVF/QCOW2/VHDX-related deployment artifacts available. Offline sensor cannot demonstrate remote polling continuity. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E14 | Reports

[Open appliance view](#)

Library includes availability, performance, inventory, alerts and headroom reports. Custom builder assembles fixed sections. No generated report history and no schedules in /reports/schedules. PDF/HTML and some Excel/CSV export controls visible; exports and delivery were not executed. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E15 | Availability

[Open appliance view](#)

One-hour fleet view explicitly said measured over 14 minutes. 54.65% availability was a transient mixed-estate sample, not a full-hour production SLA result. Coverage disclosure exists but operational acceptance remains unproven. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

E16 | Access controls

[Open appliance view](#)

Administrator and viewer roles visible; viewer visibility limited to a tag. Roles/permissions and authentication tabs present. No separate-account isolation or login test was performed. Authentication tab explicitly showed LDAP/AD and RADIUS not configured; only local sign-in was active.

E17 | Capacity view

[Open appliance view](#)

95th-percentile utilization compared with interface speed. No traffic in selected window. This is a headroom/billing view, not an exhaustion-date forecast. Development context: this snapshot is not evidence of a production failure or missing functionality; configuration, fixture and implementation findings are separated in Revision 2.

8. Source corroboration register

Local checkout: C:/Users/user/Documents/ZenPlus | HEAD b4256ff | version label 1.23.10. These source paths are references for engineering follow-up. Deployed binary equivalence was not verified.

C01 | SNMP collector and counter handling

poller/internal/checker/snmp/collector.go

High-capacity interface OIDs at lines 826-829; rate wrap/reset logic around 1094. Vendor collection and standard polling exist. No loaded-link accuracy test executed.

C02 | Standard network tables

poller/internal/checker/snmp/standard_network_groups.go

BGP at line 23, OSPF 33, VRRP/VRRPv3 42/53, STP 65/75, LACP 85. These are collector capabilities, not proof of a complete routing dashboard.

C03 | Network alert evaluator

server/app/services/network_alert_service.py

Supported canonical metric set at 43-55; template evaluation 204-246; selection/evaluation loop 468-566; apply 569-613. Compound fields are fetched but not used in the reviewed evaluation branches. Template conditions use latest values, not continuous-duration proof.

C04 | Rule API contract

server/app/api/v1/alert_rules.py

Canonical fan/PSU/VPN/HA/BGP keys accepted at 28-30; conditions mirror the first condition to legacy fields at 386-394. Compare against C03 before claiming evaluator parity.

C05 | Dependency suppression

server/app/api/v1/alert_engine.py

Upstream lookup at 381; suppression in event-driven paths around 707-843 and 1181-1278. This is not evidence of equivalent suppression in periodic SNMP evaluation.

C06 | NetPath collector / API

poller/internal/checker/netpath/prober.go

IPv4-only rejection at 109. Related API server/app/api/v1/netpath.py exposes snapshots, graph, hops, routes, events and comparison. Remote probe parity not demonstrated.

C07 | Discovery schema

server/app/schemas/discovery_v2.py

Scope/import/schedule types 12-16; protocols and credential references 33-38; classifications and limits 39-56. Local implementation corroborates empty live workflows.

C08 | Trap protocol boundary

poller/internal/checker/snmp/traps.go

Lines 39-40 explicitly defer v3 traps; current listener is v1/v2c. SNMPv3 polling and SNMPv3 trap reception are separate capabilities.

C09 | Map assistance

server/app/api/v1/manual_maps.py

LLDP/CDP link assistance begins at 603; uses topology_links around 654. Existing canvas links and auto-discovered topology must not be treated as identical lifecycle behavior.

C10 | Reporting capacity sections

server/app/services/report_sections.py

Capacity sections around 1209-1250 calculate present filesystem/link headroom. Template at 1531-1535 selects these sections. No forecast implementation found in scoped searches.

C11 | Vendor coverage migrations

scripts/migrate-080-snmp-network-device-coverage.sql

Adds CX/OS10 and extends Cisco, FortiGate, PAN-OS and MikroTik. Earlier 062/063 migrations define the main packs. Presence in source does not certify every device/firmware combination.

C12 | Authentication and role foundation

server/app/services/external_auth.py

LDAP/RADIUS and group-to-role mapping implemented; server/app/models/role.py stores permission lists. SAML/OIDC and enforcement coverage were not established.

C13 | Capture foundation

server/app/services/network_capture_service.py

Bounded on-demand capture lifecycle; agent implementation under ZenPlus_Agent/internal/netcapture. This does not establish continuous SPAN-based application/network response-time analytics.

9. Official comparator sources

All sources were consulted on 10 September 2026. Product-specific requirements, supported models and license restrictions in the linked pages govern their applicability. No third-party review is used as a technical authority.

Reference	Official documentation (click title to open)
S01	NPM product capabilities
S02	NPM administrator guide and feature index
S03	NPM documentation / current release index
S04	NPM 2026.2 release notes
S05	Polling methods
S06	Universal Device Pollers
S07	Transform poller results
S08	Wireless infrastructure and supported controller families
S09	Wireless heat-map requirements
S10	Aruba Central API monitoring
S11	Nexus monitoring requirements
S12	F5 Network Insight requirements
S13	F5 events, alerts and reports
S14	VeloCloud SD-WAN monitoring
S15	Azure gateway and site-to-site monitoring
S16	Multicast monitoring
S17	Routing Insights and its license boundary
S18	Routing neighbor monitoring and its license boundary
S19	NPM NetPath and PerfStack troubleshooting
S20	Capacity forecasting
S21	Dynamic baselines
S22	Per-object and sustained thresholds
S23	Log Viewer included with NPM
S24	Quality of Experience
S25	SPAN / mirror-port packet analysis sensors
S26	Scalability engine guidelines
S27	NPM advanced capabilities and shared platform
S28	Network Automation Manager module boundaries
S29	Network Configuration Manager scope